

Acids, Bases and Salts

Science • Chapter 2 • Chapter Summary & Study Guide for Daksh

If you have ever tasted a lemon and felt the tongue go sharp, or used baking soda to relieve indigestion, you have already met two great families of chemicals — **acids** and **bases**. This chapter explores how to identify them, what they react with, what holds them together (the H^+ and OH^- ions), and what happens when they neutralise each other to give a third family — **salts**. The chapter also brings out the role of pH, and the stories of common chemicals like baking soda, washing soda, plaster of Paris and bleaching powder.

Acids and Bases — recognising them

Indicators are dyes that change colour to tell us whether a solution is acidic or basic. Common ones:

- **Litmus** — blue in base, red in acid (a natural dye from lichen).
- **Phenolphthalein** — colourless in acid, pink in base.
- **Methyl orange** — red in acid, yellow in base.
- **Olfactory indicators** like onion or vanilla extract — their smell changes/disappears in acid or base.

All **acids** share common chemical behaviour because they ionise in water to release H^+ ions (more accurately, hydronium H_3O^+). All **bases** ionise in water to release OH^- ions. Bases that dissolve in water are called **alkalis**.

How acids and bases react

- **Acid + Metal** → **Salt + H_2** ↑. e.g. $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$. Test with a burning splinter — pop sound.
- **Acid + Metal carbonate / hydrogen carbonate** → **Salt + H_2O + CO_2** ↑. e.g. $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$. Test with lime water — turns milky.
- **Acid + Base** → **Salt + Water** (Neutralisation). e.g. $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$.
- **Base + Non-metal oxide** → **Salt + Water**. e.g. $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$. (Non-metal oxides are acidic.)
- **Acid + Metal oxide** → **Salt + Water**. e.g. $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$. (Metal oxides are basic.)

Strong vs weak acids/bases differ in how completely they ionise. HCl , H_2SO_4 , HNO_3 are strong; CH_3COOH , H_2CO_3 are weak. NaOH , KOH are strong bases; NH_4OH is a weak base. **Diluting** an acid (or base) is exothermic, so always add acid to water — never the other way round — to avoid splashing.

The pH scale

The **pH scale** measures the concentration of H^+ ions; it runs from 0 (most acidic) to 14 (most basic), with 7 being neutral. A change of 1 unit means a tenfold change in H^+ . Universal indicator paper gives the pH by colour. pH matters in everyday life:

- **Body:** stomach pH ~2 (HCl helps digestion); blood is buffered around 7.4.
- **Plants:** grow well in soil pH near 7; very acidic soils need lime.
- **Teeth:** enamel (Ca-phosphate) starts dissolving below pH 5.5 — bacteria in food residue produce acid → tooth decay. Toothpastes are mildly basic.
- **Stings:** bee/ant stings are acidic (formic acid) — relieve with mild base (baking soda); wasp sting is alkaline — relieve with mild acid like vinegar.
- **Acid rain** (pH < 5.6) damages aquatic life and corrodes monuments.

Common salts and their chemistry

Common salt (NaCl, sea / rock salt)

|

+---> Electrolysis of brine (Chlor-alkali process)
 gives NaOH (cathode) + Cl₂ (anode) + H₂ (by-prod.)

|

+---> NaOH + Cl₂ -> Bleaching powder (CaOCl₂)

|

+---> NaCl + H₂O + CO₂ + NH₃ -> NaHCO₃ -> heated -> Na₂CO₃ (washing soda)

- **Sodium hydroxide (NaOH)** — soap, paper, detergents, drain cleaner.
- **Bleaching powder (CaOCl₂)** — bleaches cotton/paper, disinfects drinking water; prepared by passing Cl₂ on dry slaked lime Ca(OH)₂.
- **Baking soda (NaHCO₃)** — antacid, baking powder ingredient, fire extinguishers. On heating it gives Na₂CO₃ + H₂O + CO₂.
- **Washing soda (Na₂CO₃·10H₂O)** — softens hard water, used in glass and paper industries.
- **Plaster of Paris (CaSO₄·½H₂O)** — got by heating gypsum (CaSO₄·2H₂O) at ~100 °C; sets hard on adding water — hence used to set fractures and make casts. Must be stored dry.
- **Water of crystallisation** — fixed water molecules in a salt's crystal (e.g. CuSO₄·5H₂O is blue; loses water on heating to give white anhydrous CuSO₄).

★ MUST-REMEMBER POINTS

- All acids in water release H⁺ ions; bases release OH⁻ ions. Without water, no ionisation, no acidic behaviour.
- Acid + metal → salt + H₂↑; Acid + carbonate → salt + H₂O + CO₂↑; Acid + base → salt + water (neutralisation).
- pH 0 (strongly acidic) — 7 (neutral) — 14 (strongly basic). Tooth enamel dissolves at pH < 5.5.
- Always add acid to water (exothermic dilution).
- Plaster of Paris formula CaSO₄·½H₂O; baking soda NaHCO₃; washing soda Na₂CO₃·10H₂O; bleaching powder CaOCl₂.
- Chlor-alkali process: 2NaCl + 2H₂O → 2NaOH + Cl₂ + H₂.

— End of Summary —